About IBM CIO TPT:

IBM CIO Technology Platform Transformation

The IBM CIO Technology Platform Transformation team plays a crucial role in modernizing and optimizing IBM's technology infrastructure and platforms. The team aims to power AI-enabled experiences through AI-first technology platforms that enable and streamline existing processes, enhance security, and improve user experience by leveraging cutting-edge technologies such as artificial intelligence, machine learning, and cloud computing.

As part of IBM's CIO TPT team, we will contribute to transformative projects that redefine enterprise operations using AI. We will directly influence productivity, user experience, and strategic decision-making across IBM.

Some of the key responsibilities of this group include:

1. Driving the adoption of emerging technologies to optimize and automate various business functions, keeping an AI-first approach with a digital experience.

2. Enable best in class IT with enhanced cybersecurity measures to protect sensitive information and maintain regulatory compliance.

3. Modernizing legacy systems and integrating disparate applications to improve interoperability and reduce technical debt.

4. Collaborating with other departments and teams to align technology efforts with broader corporate objectives.

5. Providing guidance and expertise on technology trends, best practices, and standards.

This team comprises professionals with diverse backgrounds in software engineering, data science, network architecture, and security. By fostering a culture of innovation and continuous improvement, the team strives to achieve its mission of making IBM the most productive company in the world.

About the role : AI Engineer / Data Scientist-AI

You will be part of a dynamic team driving the future of AI Deployment in large-scale enterprise transformations. We offer a collaborative environment where your technical expertise will be valued, and your professional development will be supported. Join us to work on challenging projects, leverage the latest technologies, and make a tangible impact on a leading organisation.

Job Responsibilities

Role Overview:

As an AI Engineer/SW Developer Intern, you will be in a unique position to combine your strategic thinking with your technical skills in AI, machine learning, and data analytics. You will apply your skills to help implement data-driven solutions that align with business goals. You will steer enterprise projects that improve decision-making, solve complex problems, and drive business growth. This role involves working with team members and stakeholders to translate data insights into actionable recommendations that deliver meaningful business impact.

Key Responsibilities:

1. **Implement AI, Data Science, and Technical Execution:**

- Support the design, implementation and optimization of AI-driven strategies per business stakeholder requirements.
- Design and implement machine learning solutions and statistical models, from problem formulation through deployment, to analyze complex datasets and generate actionable insights.
- Apply GenAI, traditional AI, ML, NLP, computer vision, or predictive analytics where applicable.
- Collect, clean, and preprocess structured and unstructured datasets.
- Help refine data-driven methodologies for transformation projects.
- Learn and utilize cloud platforms to ensure the scalability of AI solutions.
- Leverage reusable assets and apply IBM standards for data science and development.
- Apply ML Ops and AI ethics.

2. **Strategic Planning & Execution**
    - Translate business requirements into technical strategies.
    - Ensure alignment to stakeholders' strategic direction and tactical needs.
    - Apply business acumen to analyze business problems and develop solutions.
    - Collaborate with stakeholders and team to prioritize work.

3. **Project Management and Delivering Business Outcomes:**
    - Manage and contribute to various stages of AI and data science projects, from data exploration to model development to solution implementation and deployment.
    - Use agile strategies to manage and execute work.
    - Monitor project timelines and help resolve technical challenges.
    - Design and implement measurement frameworks to benchmark AI solutions, quantifying business impact through KPIs.

4. **Communication and Collaboration:**
    - Communicate regularly and present findings to collaborators and stakeholders, including technical and non-technical audiences.
    - Create compelling data visualizations and dashboards.
    - Work with data engineers, software developers, and other team members to integrate AI solutions into existing systems.

Qualifications:

Required Education: Pursuing a Bachelor's/ master's degree in computer science, data science, or AI, or a related field.
Preferred Education: Pursuing a Bachelor's/ master's degree in computer science, Data Science, or AI, or a related field.

Experience::
Hands-on Experience with AI/ML technologies and statistical modelling through coursework, projects, or past internships or full-time positions. Participation in AI/Data-related summits will be an added advantage (eg. Kaggle/Hackathons)

- Experience with prompt engineering or fine-tuning LLMs.
- Familiarity with tools like Lang Chain, Hugging Face Transformers, or OpenAI APIs.
- Understanding of model evaluation metrics specific to LLMs

Technical Skills:

- Proficiency in SQL and Python for performing data analysis and developing machine learning models.
- Experience and/or coursework in statistics, machine learning, generative and traditional AI.
- Knowledge of common machine learning algorithms and frameworks: linear regression, decision trees, random forests, gradient boosting (e.g., XGBoost, LightGBM), neural networks, and deep learning frameworks such as TensorFlow and PyTorch.
- Familiarity with cloud-based platforms and data processing frameworks.
- Understanding of large language models (LLMs).
- Familiarity with object-oriented programming.
- Experience and/or coursework with common Python libraries used by data scientists (e.g., NumPy, Pandas, SciPy, scikit-learn, matplotlib, Seaborn, etc.)
- Knowledge of APIs, Docker, Flask, or model serving technologies
- Experience with tools like Jupyter, Git, or cloud platforms (AWS, Azure, IBM Cloud)

Strategic and Analytical Skills:

- Strategic thinking and business acumen.
- Strong problem-solving abilities and eagerness to learn.
- Ability to work with datasets and derive insights.
- Attention to detail.

Communications and Soft Skills:

- Excellent communication skills, with the ability to explain technical concepts clearly.
- Independent and team oriented.
- Understands AI Ethics principles.
- Works openly and inclusively.
- Adaptable to fast-paced environments.
- Enthusiasm for learning and applying new technologies.
- Growth mindset.
- Ability to balance multiple initiatives, prioritize tasks effectively, and meet deadlines in a fast-paced environment.